

Strict-Singularity Rigidity for Composition Operators $H^p \rightarrow H^q$ when $q < p$

Automatic Math Discovery Smoke Test

July 19, 2026

Source Problem

Laitila, Nieminen, Saksman and Tylli ask whether their rigidity theorems for composition operators on one Hardy space have analogues for $C_\phi \in \mathcal{L}(H^p, H^q)$ when $p \neq q$.

Furthermore, the results of section 3 (in particular, see Example 3.4 and (3.3)) imply that for $p \neq 2$ there are composition operators $\widetilde{C}_\phi \in \mathcal{S}_2(h^p)$ which fail to be compact.

4.3. Open problems. Our results suggest several natural questions.

Problems 4.3. (1) Are there results corresponding to our main theorems for $C_\phi \in \mathcal{L}(H^p, H^q)$ in the case $p \neq q$? Note that the conditions for boundedness and compactness of $C_\phi : H^p \rightarrow H^q$ are different in the respective cases $p < q$ and $q < p$, and they can be found in [30], [15] and [36]. For instance, for $q < p$ one has that $C_\phi \in \mathcal{K}(H^p, H^q)$ if and only if (1.3) holds. On the other hand, the class $\mathcal{S}(L^p, L^q)$ also behaves differently from (1.4) and (1.5) for $p \neq q$. For instance, if $p, q > 2$ and $p \neq q$, then $\mathcal{S}(L^p, L^q) = \mathcal{S}_2(L^p, L^q)$ but $\mathcal{S}_p(L^p, L^q) = \mathcal{L}(L^p, L^q)$. These equalities follow from the Kadec-Pelczynski dichotomy [1, 6.4.8] and the total incomparability of ℓ^p and ℓ^q .

(2) Is there an analogue of Theorem 1.5 for $p = 1$?

(3) Is the converse of Proposition 4.1.(i) also true?

(4) Is there a Banach space E of scalar-valued analytic functions on $\mathbb{D}$ and an analytic map $\phi : \mathbb{D} \rightarrow \mathbb{D}$, for which $C_\phi \in \mathcal{S}(E) \setminus \mathcal{K}(E)$? In this direction Lefevre et. al. [19] found a non-reflexive Hardy-Orlicz space H^ψ so that $C_\phi \in \mathcal{W}(H^\psi) \setminus \mathcal{K}(H^\psi)$, where ϕ is a lens map.

The approach sketched in Subsection 4.2 suggests that weaker rigidity properties such as (4.1) are likely to hold for many other concrete classes of operators on H^p . Subsequently Miihkinen [28] has used similar techniques as in section 2 to show that the dichotomy of Theorem 1.2 remains valid for the class of analytic Volterra operators T_g on H^p , where

$$f \mapsto (T_g(f))(z) = \int_0^z f(\tau)g'(\tau)d\tau, \quad z \in \mathbb{D}.$$

We refer e.g. to the surveys [2] or [35] for the conditions on the fixed analytic map $g : \mathbb{D} \rightarrow \mathbb{C}$ which characterise the boundedness or compactness of T_g .

REFERENCES

- [1] F. Albiac and N.J. Kalton: *Topics in Banach Space Theory*, Graduate Texts in Mathematics 233 (Springer-Verlag, 2006)
- [2] A. Aleman: A class of integral operators on spaces of analytic functions, in *Topics in Complex Analysis and Operator Theory*, D. Girela and C. Gonzalez (eds.), Antiguera, 2007 (Universidad de Malaga), 3–30.
- [3] R. Aron, P. Domanski and M. Lindström: Compact homomorphisms between algebras of analytic functions, *Studia Math.* 123 (1997), 235–247.
- [4] J. Bonet, P. Domanski and M. Lindström: Essential norm and weak compactness of composition operators on weighted Banach spaces of analytic functions, *Canad. Math. Bull.* 42 (1999), 139–148.
- [5] P.S. Bourdon, J.A. Cima and A.L. Matheson: Compact composition operators on *BMOA*, *Trans. Amer. Math. Soc.* 351 (1999), 2183–2196.

The source paper notes that, in the case $q < p$, compactness of $C_\phi : H^p \rightarrow H^q$ is equivalent to

$$m(E_\phi) = 0,$$

where

$$E_\phi = \{e^{it} \in \mathbb{T} : |\phi(e^{it})| = 1\}.$$

It cites Riedl, Jarchow and Smith for the boundedness/compactness theory between different Hardy exponents.

Result

This packet records the corresponding strict-singularity rigidity in the “integrability-losing” range $q < p$.

Proof Intuition

The noncompact case $q < p$ is exactly the case in which the symbol has a positive-measure boundary contact set. The source paper already shows that positive boundary contact forces a lacunary sequence of monomials z^{n_k} whose images ϕ^{n_k} dominate an ℓ^2 basis in every finite Hardy norm. Paley’s theorem says that the original lacunary monomials are also an ℓ^2 basis in every H^p . Combining these two facts gives an ℓ^2 subspace of the domain on which $C_\phi : H^p \rightarrow H^q$ is bounded below. Thus noncompactness is incompatible with strict singularity.

Theorem 1. *Let $1 \leq q < p < \infty$, and let $\phi : \mathbb{D} \rightarrow \mathbb{D}$ be analytic. Then the following conditions are equivalent for the composition operator $C_\phi : H^p \rightarrow H^q$:*

1. C_ϕ is compact;
2. C_ϕ is strictly singular;
3. C_ϕ is ℓ^2 -singular;
4. $m(E_\phi) = 0$.

Consequently, if $C_\phi : H^p \rightarrow H^q$ is noncompact, then it fixes an isomorphic copy of ℓ^2 .

Proof. Since $q < p$ and the circle has normalized measure, the inclusion $H^p \subset H^q$ is continuous. Littlewood subordination gives boundedness of C_ϕ on H^p , so $C_\phi : H^p \rightarrow H^q$ is bounded for every analytic self-map ϕ .

Compact operators are strictly singular, and strictly singular operators are ℓ^2 -singular. Thus

$$(1) \Rightarrow (2) \Rightarrow (3).$$

The source paper records the known compactness criterion for $q < p$: one has

$$C_\phi \in \mathcal{K}(H^p, H^q) \iff m(E_\phi) = 0.$$

Hence (4) $\Rightarrow$ (1), and it remains only to show that failure of (4) implies failure of (3).

Assume $m(E_\phi) > 0$. The lacunary-power argument in Proposition 3.2 of the source paper applies with the target exponent q . Therefore there is an increasing lacunary sequence of integers

$$0 \leq n_1 < n_2 < \dots$$

and a constant $A > 0$ such that

$$\left\| \sum_k c_k \phi^{n_k} \right\|_{H^q} \geq A \|(c_k)\|_{\ell^2}$$

for every finitely supported scalar sequence (c_k) . For completeness, the mechanism is as follows. Lemma 3.1 of the source paper first gives an H^2 lower estimate for a subsequence of the powers ϕ^n on the positive-measure contact set. If $q \geq 2$, the normalized-measure inequality $\|f\|_q \geq \|f\|_2$ gives the desired lower bound. If $1 \leq q < 2$, the source proof uses the H^1 –BMOA duality estimate from Proposition 3.2 to get an H^1 lower bound, and then $\|f\|_q \geq \|f\|_1$ gives the H^q lower bound.

By passing to a further subsequence if necessary, the same sequence (n_k) may be assumed lacunary. Paley’s theorem for lacunary monomials gives a constant $B \geq 1$ such that

$$B^{-1} \|(c_k)\|_{\ell^2} \leq \left\| \sum_k c_k z^{n_k} \right\|_{H^p} \leq B \|(c_k)\|_{\ell^2}$$

for every finitely supported (c_k) . Put

$$M = \overline{\text{span}}\{z^{n_k} : k \geq 1\} \subset H^p.$$

Then M is isomorphic to ℓ^2 , and for every finitely supported (c_k) ,

$$\left\| C_\phi \left(\sum_k c_k z^{n_k} \right) \right\|_{H^q} = \left\| \sum_k c_k \phi^{n_k} \right\|_{H^q} \geq A \|(c_k)\|_{\ell^2} \geq \frac{A}{B} \left\| \sum_k c_k z^{n_k} \right\|_{H^p}.$$

By density, the same lower estimate holds on all of M . Hence C_ϕ is bounded below on a subspace of H^p isomorphic to ℓ^2 , so C_ϕ is not ℓ^2 -singular. This proves (3) $\Rightarrow$ (4) by contrapositive, and all four conditions are equivalent. $\square$

Corollary 1. *For $1 \leq q < p < \infty$, every noncompact composition operator $C_\phi : H^p \rightarrow H^q$ is non-strictly singular. In fact it fixes a copy of ℓ^2 .*

Scope and Novelty Check

This is a partial answer to Problem 4.3(1) of the source paper. It settles the finite-exponent range $q < p$, where compactness is governed by the boundary contact condition. It does not address the opposite range $p < q$, where boundedness itself is a stronger Carleson-type condition and the source’s lacunary monomial span does not by itself give a bounded-below map from an ℓ^p domain model into an ℓ^q image model.

Cheap run indexes were searched for 1607.00113, the source title, `H^p, H^q, Hardy, composition operator`, and `strict singularity`. No exact prior local packet for this cross-exponent partial result was found. Bounded web checks for `strictly singular composition operators H^p H^q`, `composition operators between Hardy spaces strict singular`, and related phrases found the source paper and adjacent weighted-composition literature, but no later result explicitly recording this $q < p$ implication.

Human Review Recommendation

Review should focus on the reuse of Proposition 3.2 of the source paper with the target exponent renamed to q , and on the cited compactness criterion for $C_\phi : H^p \rightarrow H^q$ in the case $q < p$. The rest is a direct Paley lacunary-subspace argument.

References

- [1] J. Laitila, P. J. Nieminen, E. Saksman and H.-O. Tylli, *Rigidity of composition operators on the Hardy space H^p* , Advances in Mathematics 319 (2017), 610–629. arXiv:1607.00113.
- [2] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970; Dover, 2000.
- [3] C. C. Cowen and B. D. MacCluer, *Composition Operators on Spaces of Analytic Functions*, CRC Press, 1995.